

Explanation

The script calculates the maximum allowable exposure time for a camera to avoid motion blur that exceeds a specified threshold. This is crucial in scenarios requiring high precision, such as optical target tracking.

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Formulation

Motion blur occurs when a target or the camera moves while the camera's shutter is open. The moving target will spread across several pixels on the sensor, resulting in blur. The amount of motion blur depends on: - Exposure Time (t_{exp}): The time the camera's sensor is exposed to light. - Relative Velocity (v_{rel}): The speed of the target relative to the camera. - Distance to Target (z): The distance between the camera and the target. - Focal Length (f): The distance between the lens and the sensor, measured in millimeters. - Pixel Size (px_size): The physical size of each pixel on the sensor, measured in millimeters.

```
% The amount of motion blur in pixels can be expressed as:
% Motion Blur = (v_rel * t_exp * f) / (z * px_size)

% To find the exposure time, the equation is rearranged as follows:
% t_exp = (blur_thresh * z * px_size) / (v_rel * f)
% This formula allows you to calculate the maximum exposure time that will keep motion blur
% within the acceptable threshold.
```

Clear previous data

```
clear;
clc;

format long G
```

CAMERA INTRINSIC PARAMETERS

Inputs

```
res_h_px = 960;    % Image height in pixels
res_w_px = 1280;   % Image width in pixels
focal_len_mm = 4;  % Focal length of the lens in millimeters
sens_w_mm = 5.7;   % Sensor width in millimeters
sens_h_mm = 4.28;  % Sensor height in millimeters

% Calculations
px_size_mm = sens_w_mm / res_w_px; % Pixel size in millimeters
c_x_px = res_w_px / 2;             % Principal point X-coordinate in pixels
c_y_px = res_h_px / 2;             % Principal point Y-coordinate in pixels
f_x_px = focal_len_mm * res_w_px / sens_w_mm; % Focal length in X direction (pixels)
f_y_px = focal_len_mm * res_h_px / sens_h_mm; % Focal length in Y direction (pixels)

% Field of View (FoV) Calculations
fov_h_deg = 2 * rad2deg(atan(sens_w_mm / (2 * focal_len_mm))); % Horizontal FoV in degrees
fov_v_deg = 2 * rad2deg(atan(sens_h_mm / (2 * focal_len_mm))); % Vertical FoV in degrees
```

TARGET AND CAMERA MOTION PARAMETERS

Inputs

```
dist_to_tgt_m = 50;    % Perpendicular distance to target in meters
tgt_w_m = 0.5;         % Target width in meters
tgt_h_m = 1.8;         % Target height in meters
tgt_vel_h_m_s = 35;    % Horizontal target speed in meters per second
tgt_vel_v_m_s = 1;     % Vertical target speed in meters per second

% Calculations
fov_w_at_dist_m = 2 * tand(fov_h_deg / 2) * dist_to_tgt_m; % FoV width at target distance in meters
fov_h_at_dist_m = 2 * tand(fov_v_deg / 2) * dist_to_tgt_m; % FoV height at target distance in meters
px_per_m_h = res_w_px / fov_w_at_dist_m; % Pixels per meter horizontally at target distance
px_per_m_v = res_h_px / fov_h_at_dist_m; % Pixels per meter vertically at target distance

tgt_px_w_at_dist = f_x_px * tgt_w_m / dist_to_tgt_m; % Target width in pixels at given distance
tgt_px_h_at_dist = f_y_px * tgt_h_m / dist_to_tgt_m; % Target height in pixels at given distance
```

MOTION BLUR PARAMETERS

Inputs

```
tgt_disp_m = 1;           % Target displacement in the world in meters
blur_thresh_px = 1;       % Allowed motion blur threshold in pixels

% Calculations
tgt_disp_px_h = f_x_px * tgt_disp_m / dist_to_tgt_m; % Displacement in pixels horizontally
tgt_disp_px_v = f_y_px * tgt_disp_m / dist_to_tgt_m; % Displacement in pixels vertically

travel_dist_px_h_m = blur_thresh_px * dist_to_tgt_m / f_x_px; % Target travel distance causing pixel change horizontally
travel_dist_px_v_m = blur_thresh_px * dist_to_tgt_m / f_y_px; % Target travel distance causing pixel change vertically

exp_time_h_s = travel_dist_px_h_m / tgt_vel_h_m_s; % Needed exposure time for horizontal motion blur threshold
exp_time_v_s = travel_dist_px_v_m / tgt_vel_v_m_s; % Needed exposure time for vertical motion blur threshold
```

RESULTS OUTPUT

```
fprintf("Target size (W, H): (%.2f, %.2f) meters\n", tgt_w_m, tgt_h_m);
fprintf("Distance to target: %.2f meters\n", dist_to_tgt_m);
fprintf("Allowed motion blur: %.2f pixels\n", blur_thresh_px);
fprintf("Motion blur as %% of target height: %.2f%% \n", (blur_thresh_px / tgt_px_h_at_dist) * 100);
fprintf("Motion blur as %% of target width: %.2f%% \n", (blur_thresh_px / tgt_px_w_at_dist) * 100);
fprintf("Target pixel size (W, H) at distance %.2f meters: (%.2f, %.2f) pixels\n\n", dist_to_tgt_m, tgt_px_w_at_dist, tgt_px_h_at_dist);

fprintf("Horizontal target speed: %.2f m/s\n", tgt_vel_h_m_s);
fprintf("Exposure time for %.2f pixel blur at %.2f meters and %.2f m/s: %f s\n", blur_thresh_px, dist_to_tgt_m, tgt_vel_h_m_s, exp_time_h_s);
fprintf("Exposure time for %.2f pixel blur at %.2f meters and %.2f m/s: %f ms\n", blur_thresh_px, dist_to_tgt_m, tgt_vel_h_m_s, exp_time_h_s * 1000);

fprintf("Vertical target speed: %.2f m/s\n", tgt_vel_v_m_s);
fprintf("Exposure time for %.2f pixel blur at %.2f meters and %.2f m/s: %f s\n", blur_thresh_px, dist_to_tgt_m, tgt_vel_v_m_s, exp_time_v_s);
fprintf("Exposure time for %.2f pixel blur at %.2f meters and %.2f m/s: %f ms\n", blur_thresh_px, dist_to_tgt_m, tgt_vel_v_m_s, exp_time_v_s * 1000);
```

```
Target size (W, H): (0.50, 1.80) meters
Distance to target: 50.00 meters
Allowed motion blur: 1.00 pixels
Motion blur as % of target height: 3.10%
Motion blur as % of target width: 11.13%
Target pixel size (W, H) at distance 50.00 meters: (8.98, 32.30) pixels

Horizontal target speed: 35.00 m/s
Exposure time for 1.00 pixel blur at 50.00 meters and 35.00 m/s: 0.001590 s
Exposure time for 1.00 pixel blur at 50.00 meters and 35.00 m/s: 1.590402 ms

Vertical target speed: 1.00 m/s
Exposure time for 1.00 pixel blur at 50.00 meters and 1.00 m/s: 0.055729 s
Exposure time for 1.00 pixel blur at 50.00 meters and 1.00 m/s: 55.729167 ms
```